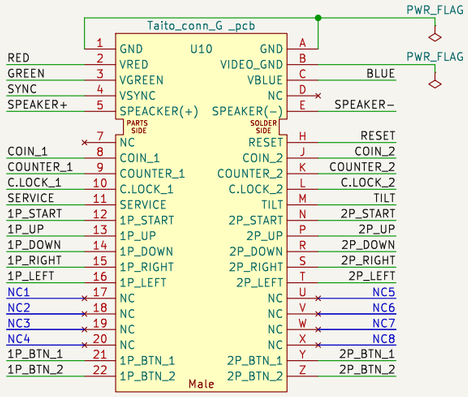
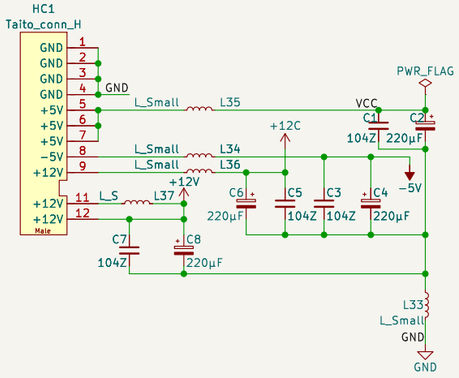
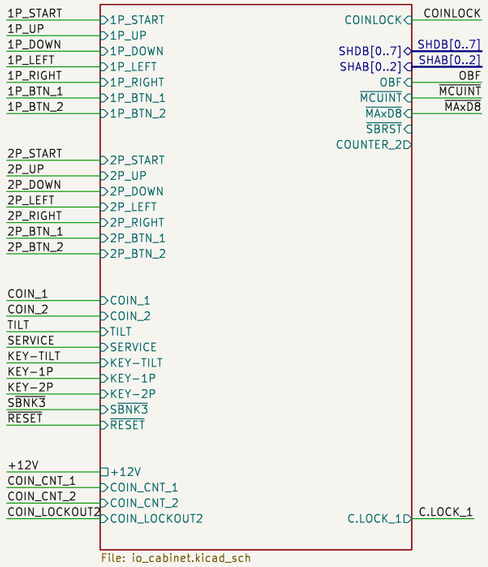


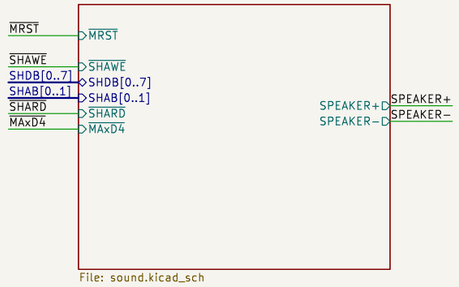




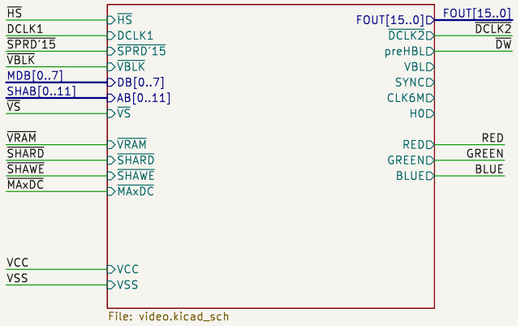
Cabinet



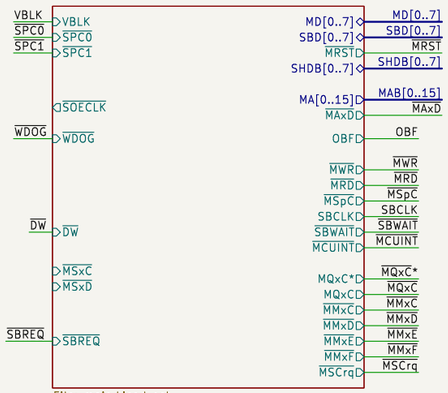
Sound



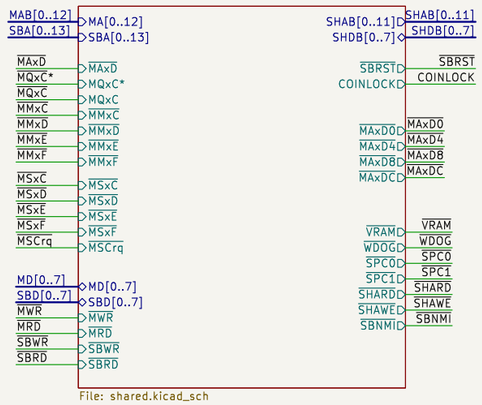
Video



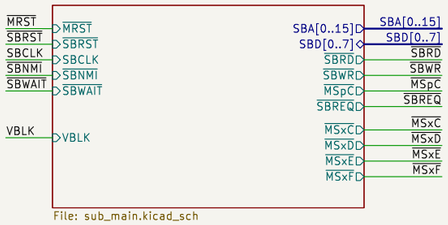
Main



Shared



Sub



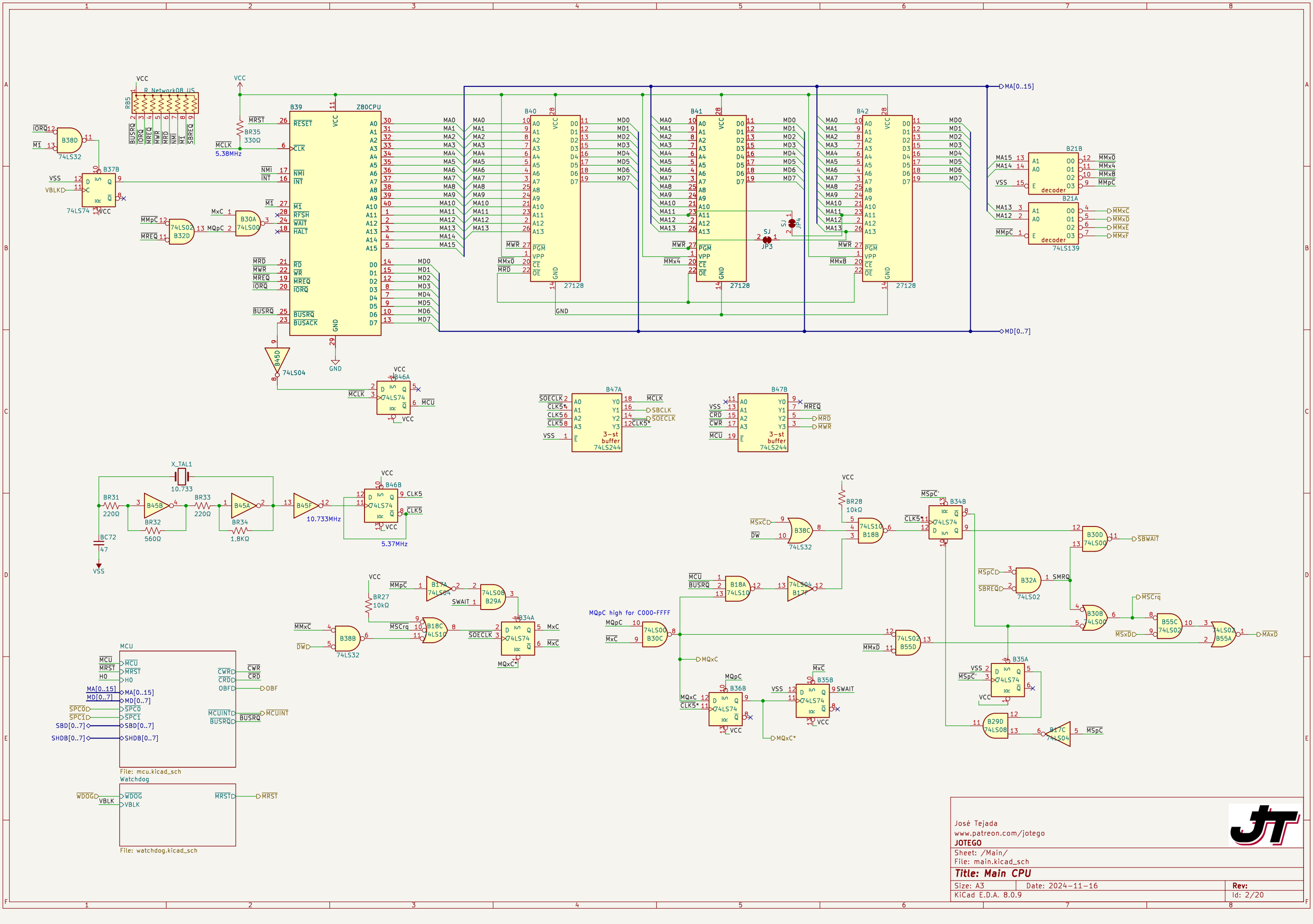


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Sheet: /
File: fairyland.kicad_sch

Title: The Fairyland Story

Size: A3	Date: 2024-11-16	Rev:
KiCad E.D.A. 8.0.9		Id: 1/20



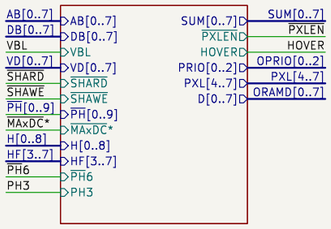
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Sheet: /Main/
File: main.kicad_sch

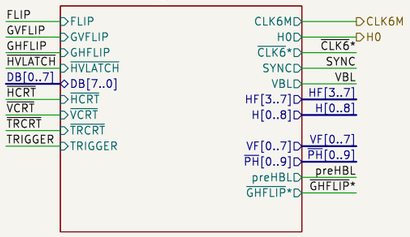
Title: Main CPU

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.9		Id: 2/20

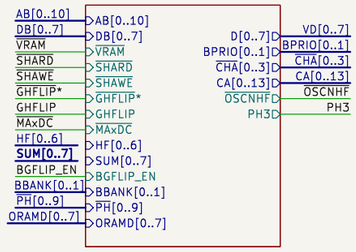
Object RAM



Timing

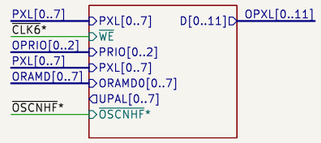


Video RAM

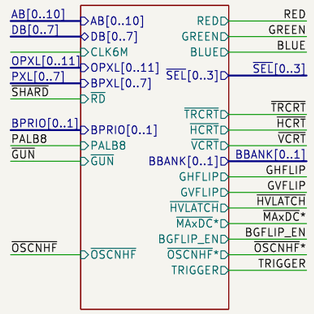


Tile ROM

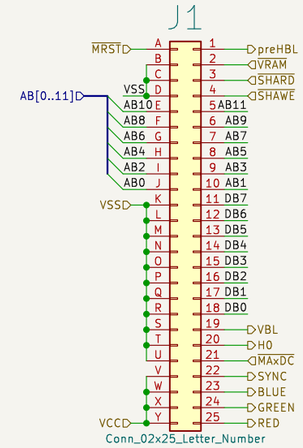
Line buffer



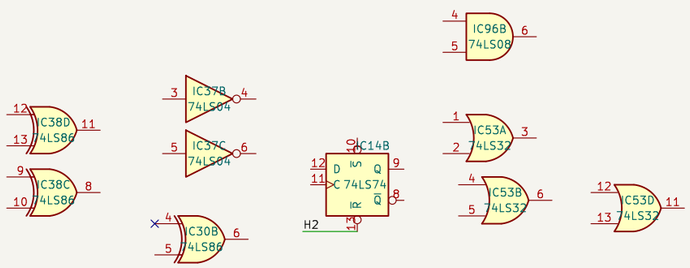
Palette



J1



Miscellaneous



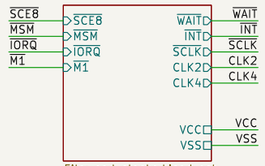
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Sheet: /Video/
File: video.kicad_sch

Title: The Fairyland Story

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.9		Id: 3/20

Clocks



File: snd_clocks.kicad_sch

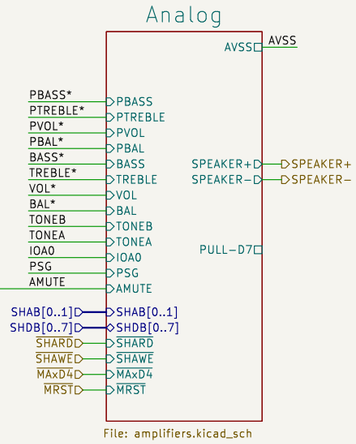
Sound Generators

File: ongen.kicad_sch

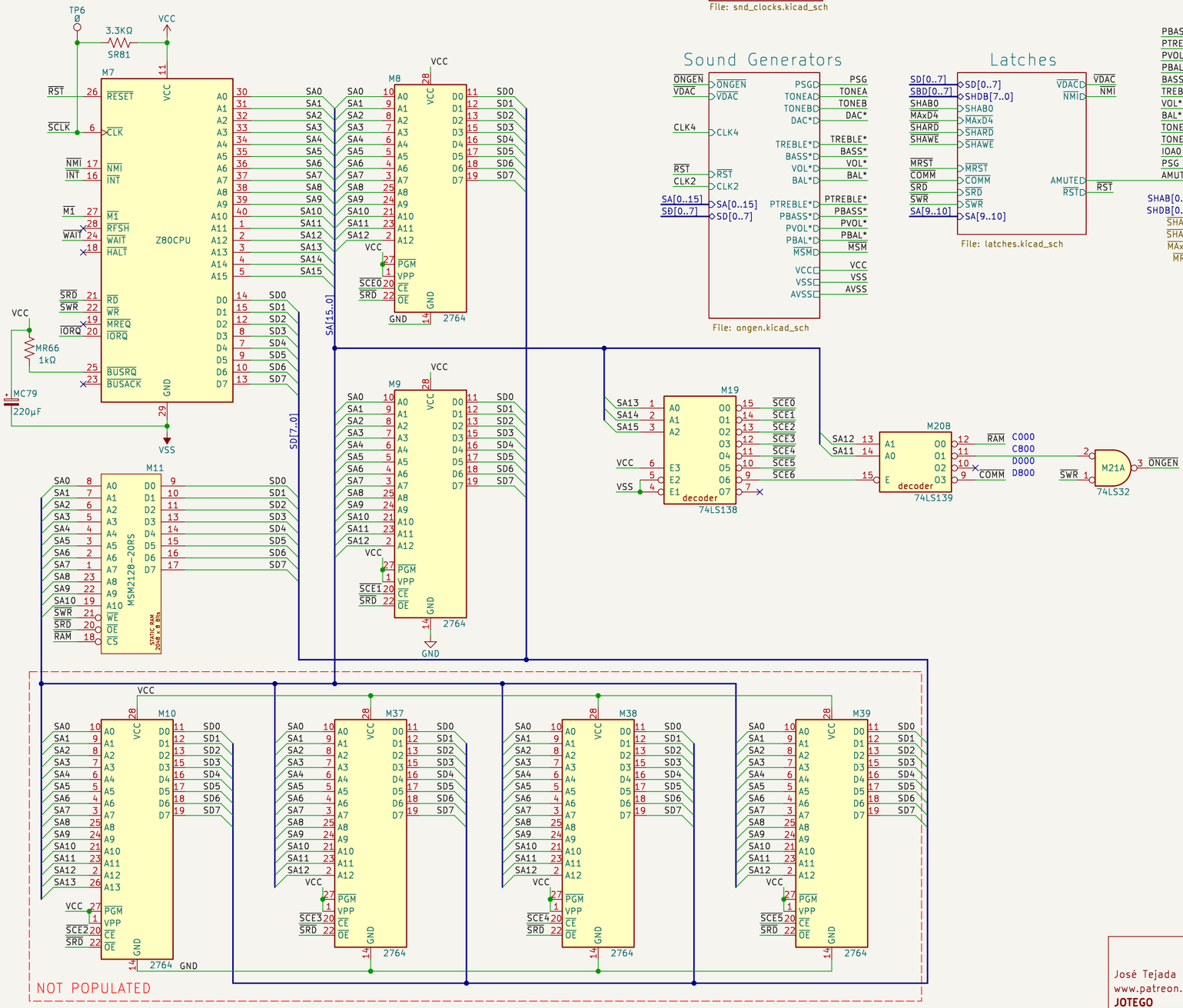
Latches

File: latches.kicad_sch

Analog



File: amplifiers.kicad_sch

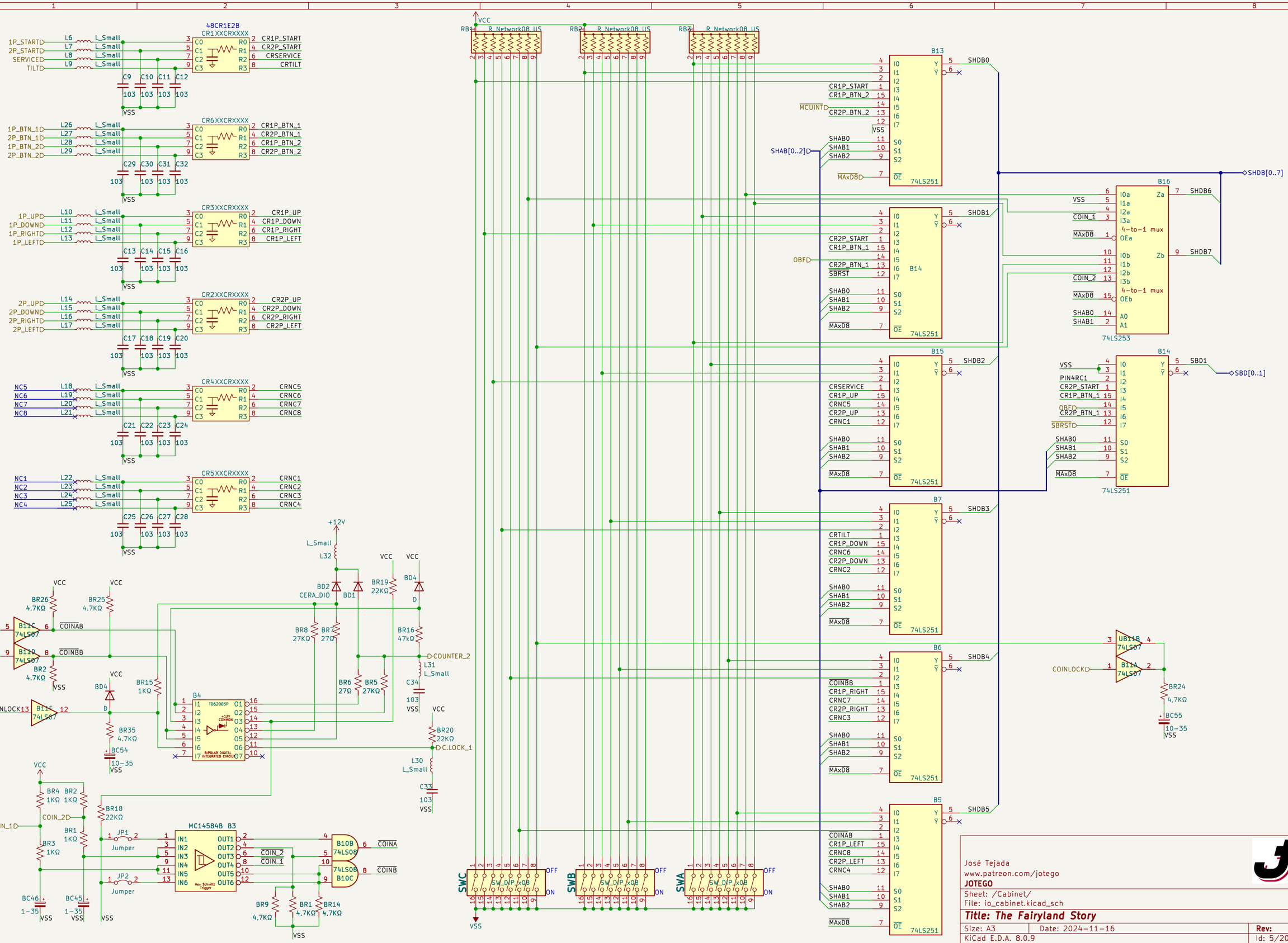


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Sheet: /Sound/
File: sound.kicad_sch

Title: Sound Board

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.9		Id: 4/20


48CR1E2B
CR1XXCRXXXX
1P_START
2P_START
SERVICE
TILT
CR1P_START
CR2P_START
CRSERVICE
CRTILT
CR6XXCRXXXX
1P_BTN_1
2P_BTN_1
1P_BTN_2
2P_BTN_2
CR1P_BTN_1
CR2P_BTN_1
CR1P_BTN_2
CR2P_BTN_2
CR3XXCRXXXX
1P_UP
1P_DOWN
1P_RIGHT
1P_LEFT
CR1P_UP
CR1P_DOWN
CR1P_RIGHT
CR1P_LEFT
CR2XXCRXXXX
2P_UP
2P_DOWN
2P_RIGHT
2P_LEFT
CR2P_UP
CR2P_DOWN
CR2P_RIGHT
CR2P_LEFT
CR4XXCRXXXX
NC5
NC6
NC7
NC8
CRNC5
CRNC6
CRNC7
CRNC8
CR5XXCRXXXX
NC1
NC2
NC3
NC4
CRNC1
CRNC2
CRNC3
CRNC4
VSS
VCC
+12V
R_Network08_US
RB1
RB2
RB3
B13
B14
B15
B7
B6
B5
74LS251
MCUINT
SHAB[0..2]
OBF
SBRST
SHDB0
SHDB1
SHDB2
SHDB3
SHDB4
SHDB5
SHDB[0..7]
B16
74LS253
4-to-1 mux
COIN_1
COIN_2
SBD1
SBD[0..1]
PIN4RC1
SWC
SWB
SWA
SW_DIP_x08
OFF
ON
B11C
B11D
B11E
B11A
B11B
74LS07
COINA
COINB
COINAB
COINBB
COINLOCK
COINLOCKD
B10B
B10C
74LS08
MC14584B B3
Hex Schmitt Trigger
JP1
JP2
Jumper
TD62003P
BIPOLAR DIGITAL INTEGRATED CIRCUIT
COMMON
CERA_DIO
COUNTER_2
C.LOCK_1
BD1
BD2
BD4
BR1
BR2
BR3
BR4
BR5
BR6
BR7
BR8
BR9
BR14
BR15
BR16
BR18
BR19
BR20
BR24
BR25
BR26
BR35
BC45
BC46
BC54
BC55
L6
L7
L8
L9
L10
L11
L12
L13
L14
L15
L16
L17
L18
L19
L20
L21
L22
L23
L24
L25
L26
L27
L28
L29
L30
L31
L32
L33
L34
C9
C10
C11
C12
C13
C14
C15
C16
C17
C18
C19
C20
C21
C22
C23
C24
C25
C26
C27
C28
C29
C30
C31
C32
C33
C34
103
4.7KΩ
22KΩ
27KΩ
47KΩ
1KΩ
27Ω
10-35
L_Small
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Sheet: /Cabinet/
File: io_cabinet.kicad_sch
Title: The Fairyland Story
Size: A3
Date: 2024-11-16
Rev:
KiCad E.D.A. 8.0.9
Id: 5/20

device select signals driven by either sub or main CPUs

PIN2 SOCKET B48 (NOT USED)

Cabinet I/O

PCB can support 4kB usage too

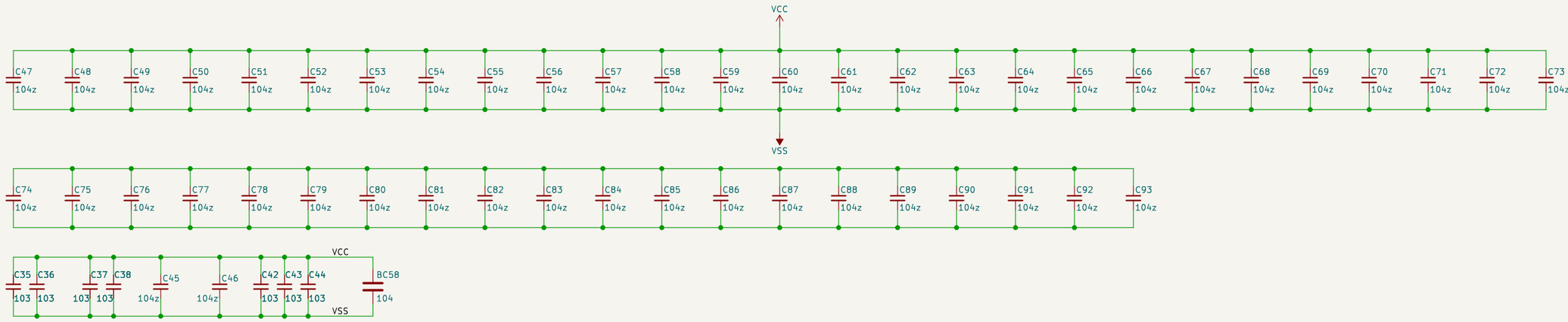
Used in NY Captor as ROM bank control

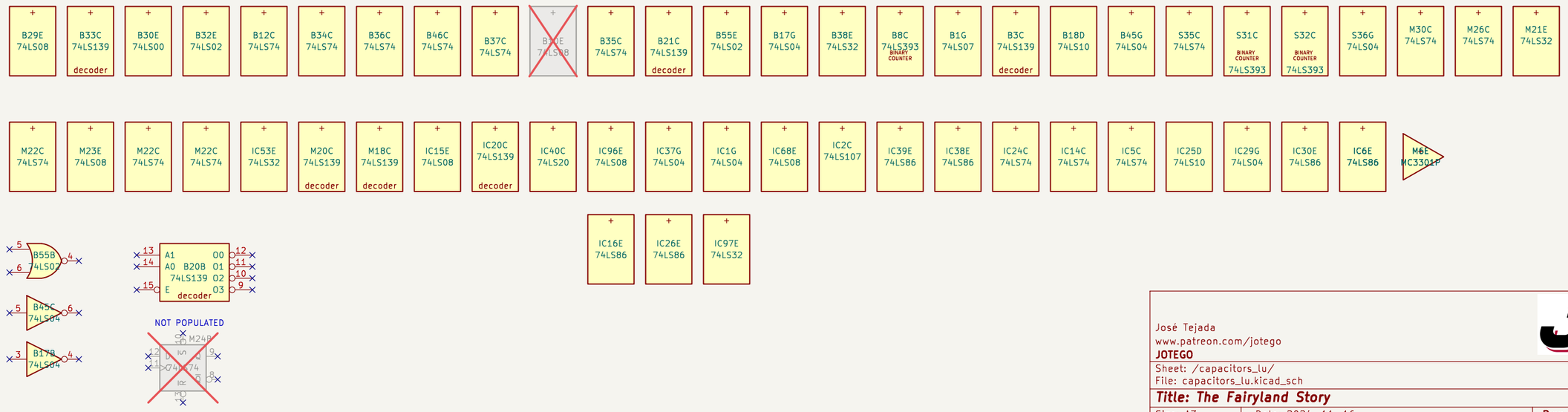
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Sheet: /Shared/
File: shared.kicad_sch

Title: Shared bus and RAM

Size: A3	Date: 2024-11-16	Rev:
KiCad E.D.A. 8.0.9		Id: 6/20



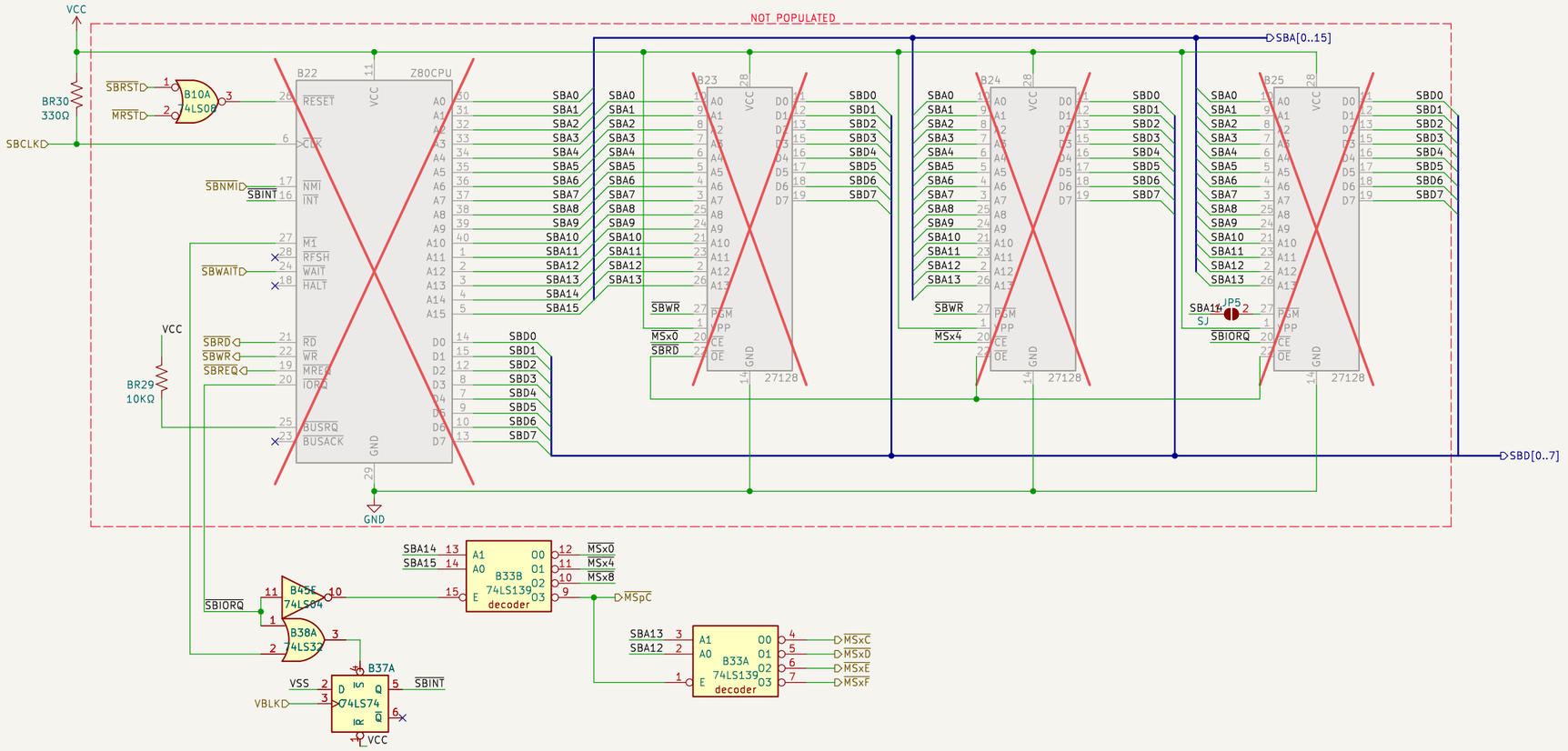


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Sheet: /capacitors_lu/
File: capacitors_lu.kicad_sch

Title: The Fairyland Story

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.9		Id: 7/20

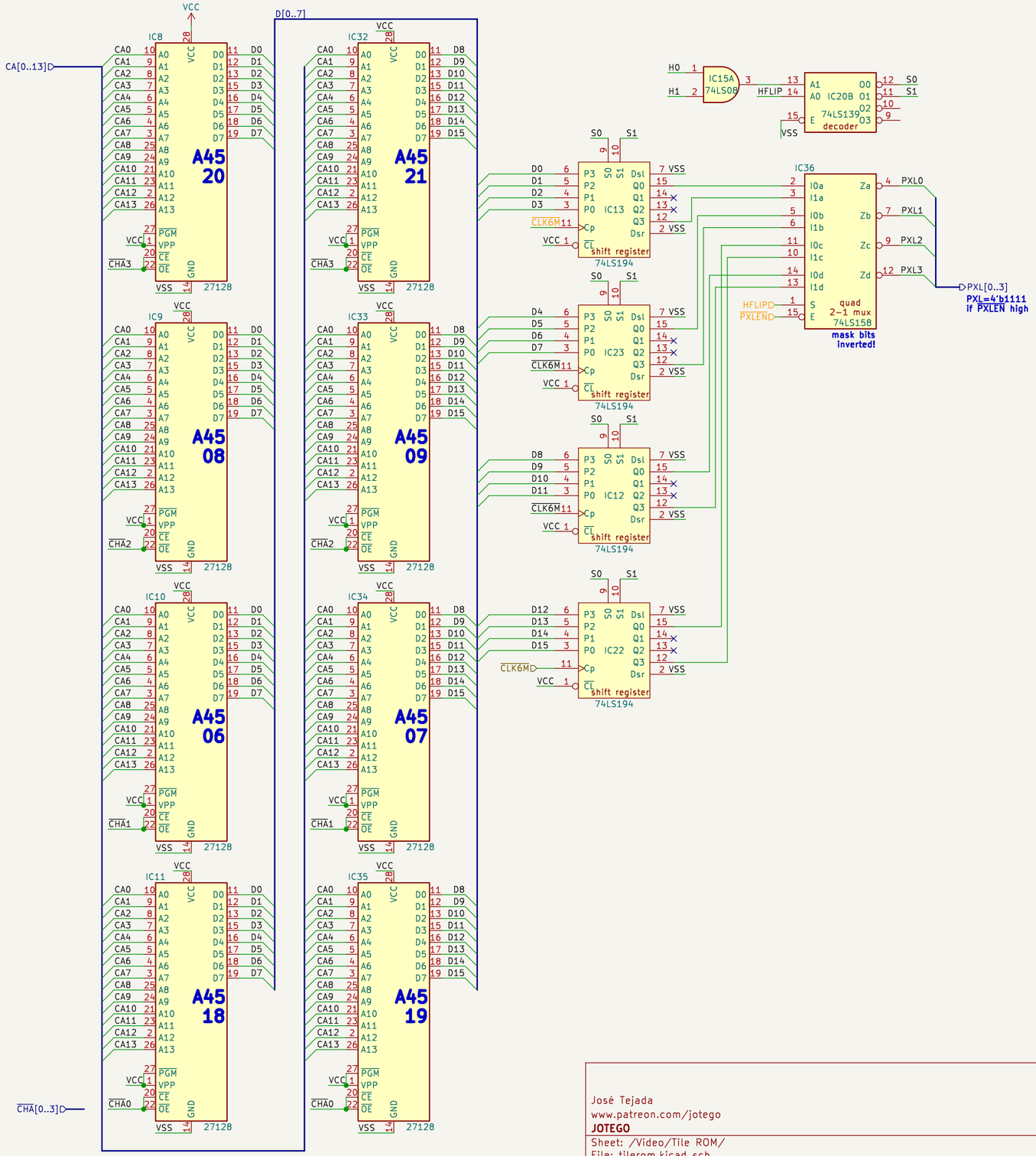


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Sheet: /Sub/
File: sub_main.kicad_sch

Title: Subsidiary CPU

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.9		Id: 8/20



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Sheet: /Video/Tile ROM/
File: tilerom.kicad_sch

Title: Tile ROMs

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.9		Id: 9/20

XTAL1
18.432MHz
BC6
20
SC7
20
IC1D
IC1F
IC1E
74LS04
BR9
330Ω
BR2
330Ω
IC2A
IC2B
74LS107
VCC
CLK6M
CLK6*
IC1A
CLK6M
6.144MHz

IC14A
GHFLIP
PH1
PH6
74LS74
GHFLIP*

IC18
IC28
74LS161
counter
VSS
VCC
H0
CLK6M
PE
CEP
CET
CP
MR
TC
H1
H2
H3
H4
H5
H6
H7
H8
H[0..8]

IC39B
IC39D
IC39C
IC39A
IC38A
74LS86
HF3
HF4
HF5
HF6
HF7
During PH6,
HF=H
HF[3..7]

IC5A
IC5B
IC1C
IC1B
74LS04

IC70
74LS42
4-LINE BCD TO 10-LINE DECIMAL DECODER
3.08MHz H0
1.54MHz H1
769kHz H2
6.25MHz CLK6M
PH0
PH1
PH2
PH3
PH4
PH5
PH6
PH7
PH8
PH9
PH[0..9]
769kHz
Each signal is active
every 8 pixels

IC29E
IC15B
IC19A
IC30D
74LS08
74LS86
H5
H6
H4
HBB
V8
SYNC

IC25A
IC25B
IC19B
HOVER
preHBL
V5
V6
V7
V4
VBL
line 240. VBL starts
line 16, VBL ends

IC17
IC27
VC
V0
V1
V2
V3
V4
V5
V6
V7
VC

IC26C
IC16A
IC16D
IC16B
IC16C
IC26A
IC26D
IC26B
VF0
VF1
VF2
VF3
VF4
VF5
VF6
VF7
GVFLIP
VF[0..7]

R_Network08_US
ICR1
DB0
DB1
DB2
DB3
DB4
DB5
DB6
DB7
DB[7..0]

IC41
IC42
74LS373
LATCHES AND FLIP FLOPS
VCRT
HCRT
IC52_19
This seems to be
a circuit to capture
when an event occurs
relative to the raster
position. It might
have been planned for a gun.

IC3A
IC3B
IC4C
74LS125
HVLATCH
GND
TRCRT

TRIGGER
L
L2
C1
221

HORIZONTAL COUNTER
from 080 ~ 1FF
blanking 080~0FF (128 pixels)
active 100~1FF (256 pixels)

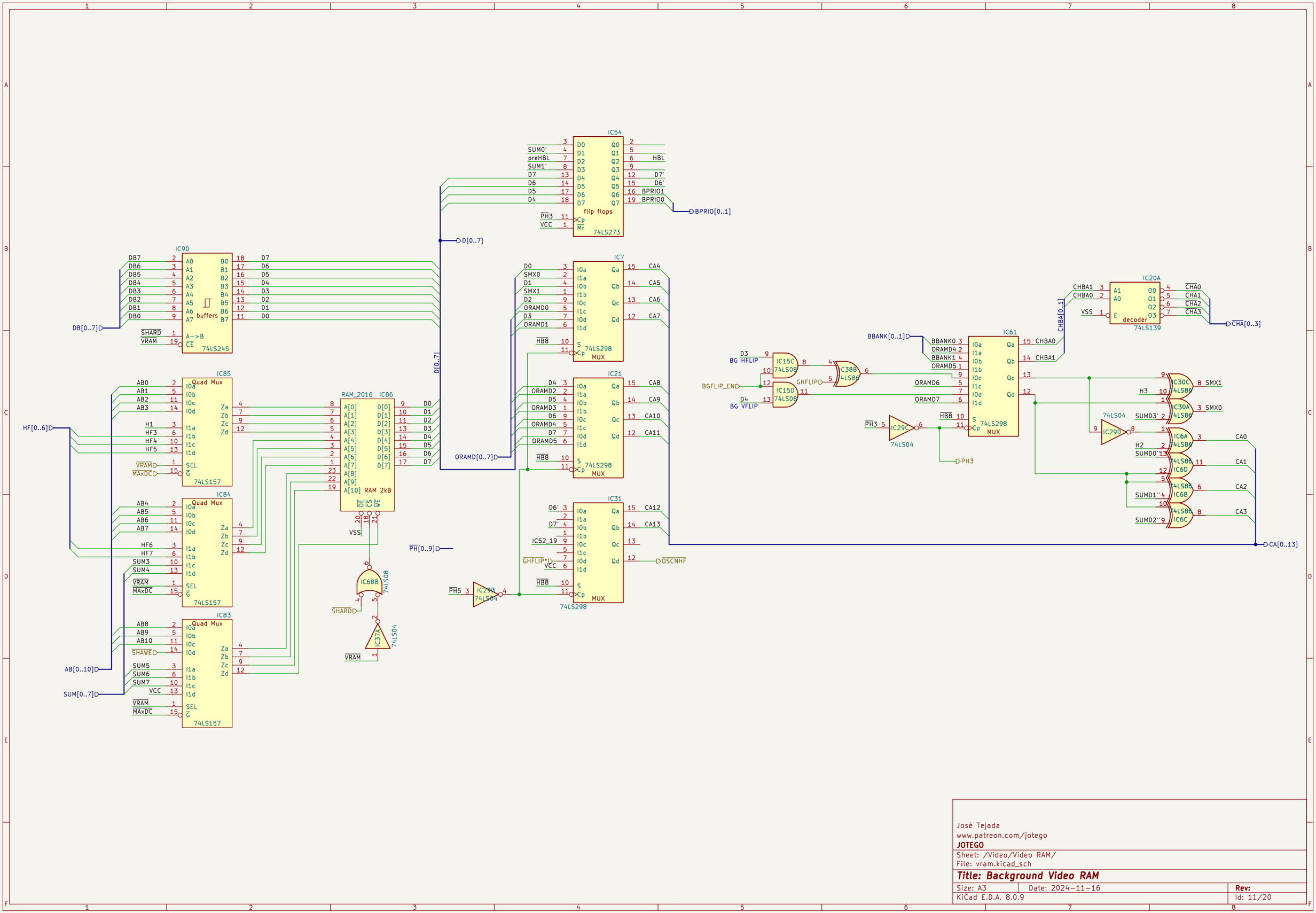
VERTICAL COUNTER
from 0F0 ~ 1FF
blanking 1F0~1FF~0F0~0FF (32 lines)
active 100~1EF (240 lines)

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Sheet: /Video/Timing/
File: timing.kicad_sch

Title: Video Timing

Size: A3 | Date: 2024-11-16 | **Rev:**
KiCad E.D.A. 8.0.9 | Id: 10/20

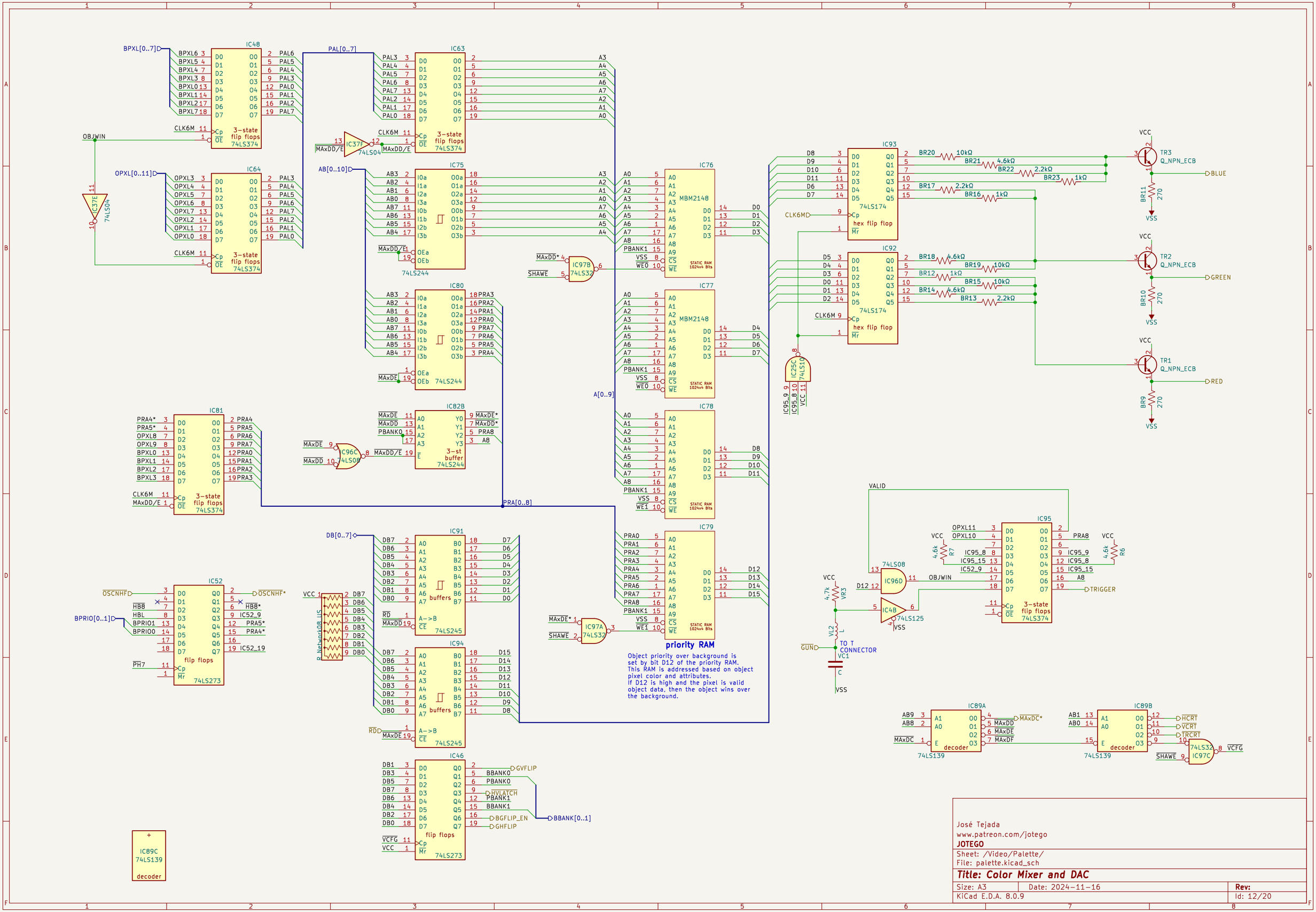


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Sheet: /Video/Video RAM/
File: vram.kicad_sch

Title: Background Video RAM

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.9		Id: 11/20

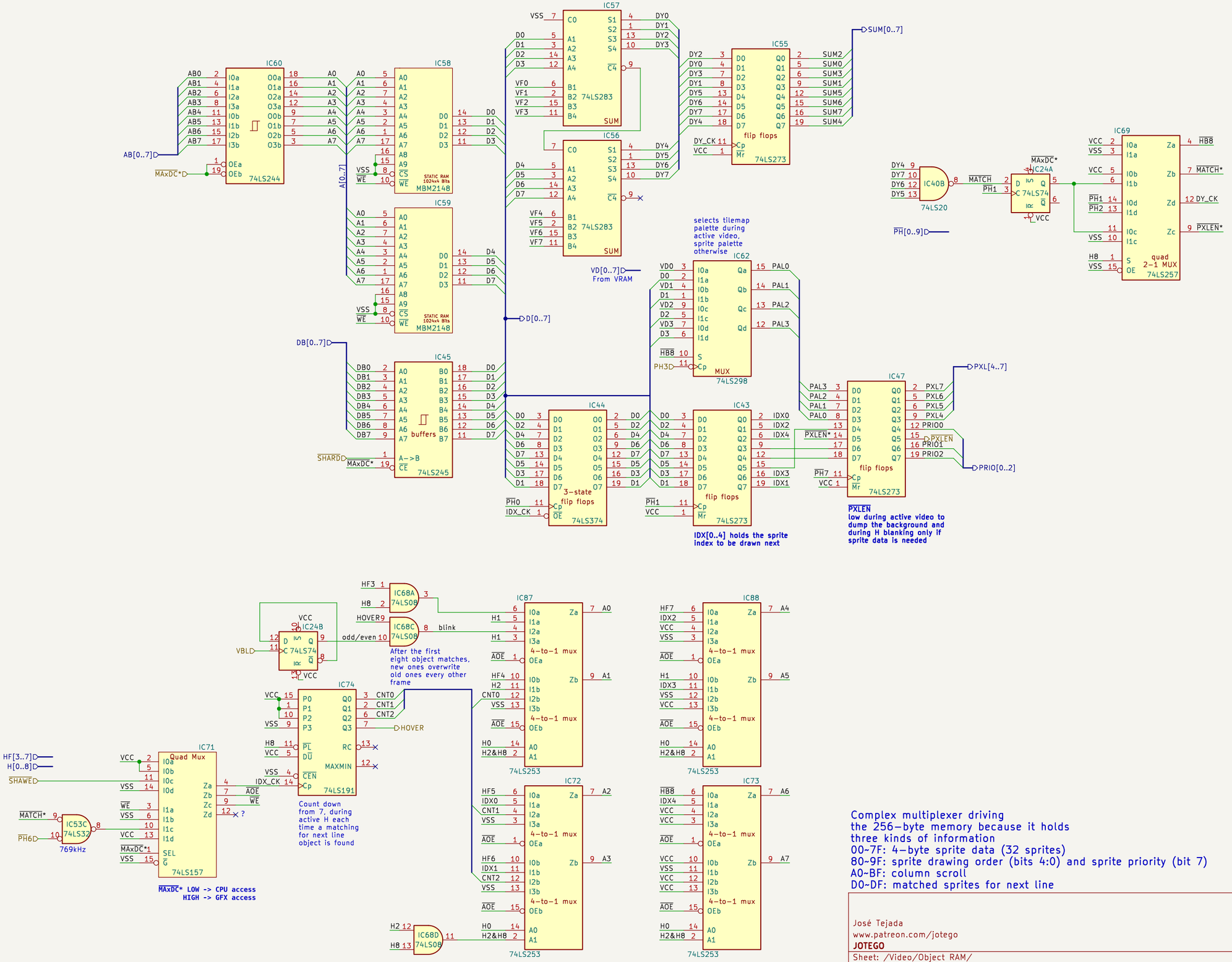


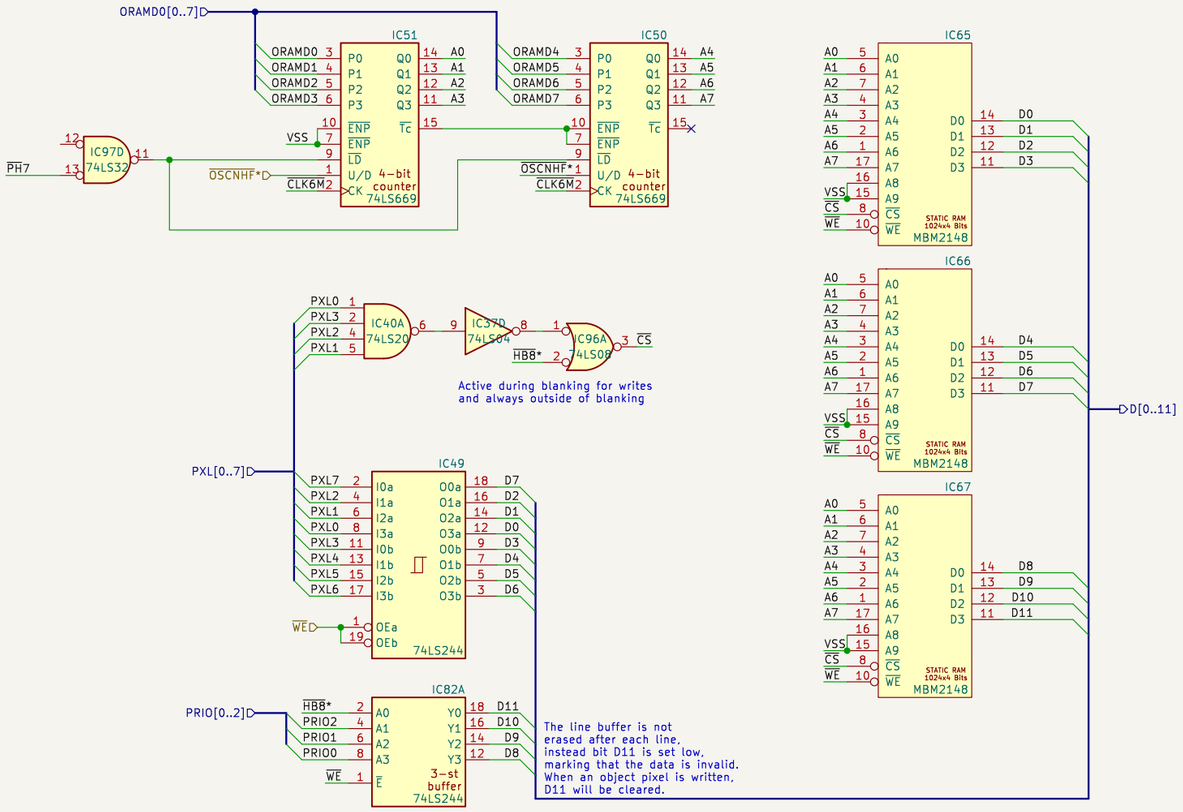
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Sheet: /Video/Palette/
File: palette.kicad_sch

Title: Color Mixer and DAC

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.9		Id: 12/20


IC60
74LS244
IC58
MBM2148
IC59
MBM2148
IC57
74LS283
IC56
74LS283
IC55
74LS273
flip flops
SUM[0..7]
IC45
buffers
74LS245
IC44
74LS374
3-state
flip ops
IC43
74LS273
flip flops
VD[0..7]
From VRAM
selects tilemap
palette during
active video,
sprite palette
otherwise
IC62
74LS298
MUX
IC47
74LS273
flip flops
PXL[4..7]
PRIO[0..2]
PXLEN
low during active video to
dump the background and
during H blanking only if
sprite data is needed
IDX[0..4] holds the sprite
index to be drawn next
IC40B
74LS20
IC24A
74LS74
IC69
74LS257
quad
2-1 MUX
IC68A
74LS08
IC68C
74LS08
IC68D
74LS08
blink
IC24B
74LS74
odd/even
After the first
eight object matches,
new ones overwrite
old ones every other
frame
IC74
74LS191
MAXMIN
Count down
from 7, during
active H each
time a matching
for next line
object is found
HOVER
IC71
Quad Mux
74LS157
IC53C
74LS32
769kHz
MAxDC* LOW -> CPU access
HIGH -> GFX access
IC87
74LS253
4-to-1 mux
IC88
74LS253
4-to-1 mux
IC72
74LS253
4-to-1 mux
IC73
74LS253
4-to-1 mux
Complex multiplexer driving
the 256-byte memory because it holds
three kinds of information
00~7F: 4-byte sprite data (32 sprites)
80~9F: sprite drawing order (bits 4:0) and sprite priority (bit 7)
A0~BF: column scroll
D0~DF: matched sprites for next line
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JOTEGO
Sheet: /Video/Object RAM/
File: obj_ram.kicad_sch
Title: Object Information
Size: A3
Date: 2024-11-16
Rev:
KiCad E.D.A. 8.0.9
Id: 13/20

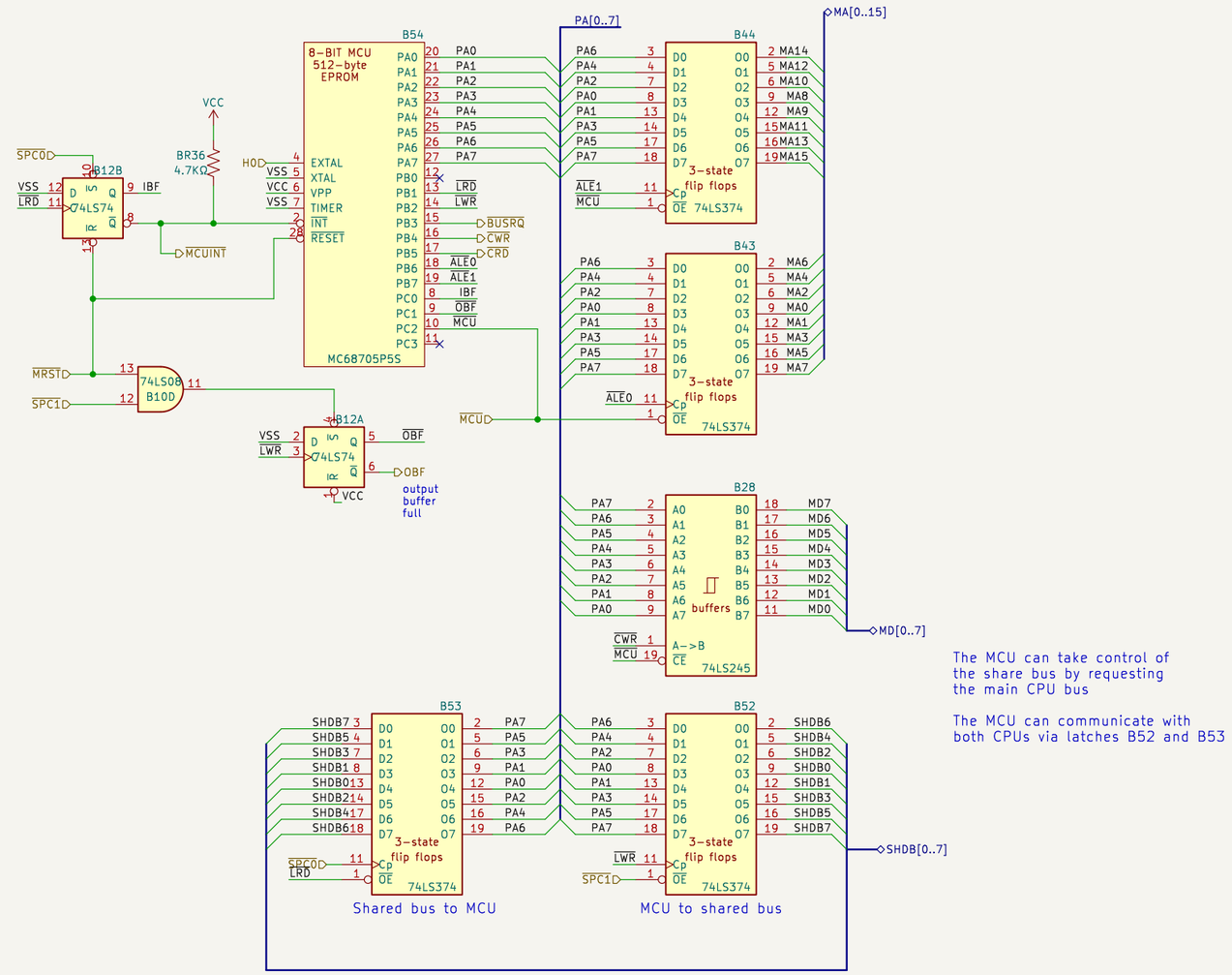


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Sheet: /Video/Line buffer/
File: linebuf.kicad_sch

Title: Single Line Video Buffer

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.9		Id: 14/20

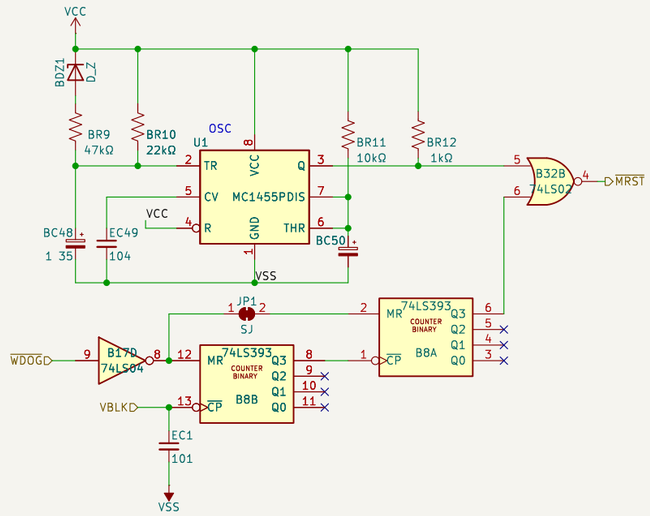


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Sheet: /Main/MCU/
File: mcu.kicad_sch

Title: MCU

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.9		Id: 15/20



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Sheet: /Main/Watchdog/
File: watchdog.kicad_sch

Title:

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.9		Id: 16/20

PULL-D7
MC40 472K
MR33 1MΩ
MR32 20MΩ
MR82 1.5kΩ
VSS
M4B MC3301P
DA TEST
MR34 20MΩ
MC41 151K
MC39 104K
MR38 4.7kΩ
+12V
MC18 332K
MC19 473K
MC20 104K
MC21 104K
MC79 104K
MC81 104K
MR11 1.5kΩ
MC24 47µF
M4E MC3301P
MR83 10kΩ
MR36 5kΩ
MC68 10µF
MR37 1MΩ
MC16 47µF
MR6 100kΩ
M2
TA7630P
DUAL VOLUME/BALANCE /TONE(BASS/TREBLE)
REF_SIG TH(2) TL(2) TL(1) TH(1) VCC
IN-2 IN-1 OUT-2 OUT-1
Vref Balance Volume Bass Treble VSS
MR14 10kΩ
MR16 10kΩ
MR10 1.5kΩ
MR15 75kΩ
MR17 75kΩ
MC26 104K
PSG
MR8 10kΩ
MC17 104K
MR9 1kΩ
MR7 10kΩ
MR85 1MΩ
OUT-1
M4D MC3301P
TEST KG
MC15 10µF
MR35 4.7kΩ
MR84 1MΩ
MR86 1MΩ
MC69 3.3
MR40 1MΩ
M4A MC3301P
MC70 3.3
MR39 1MΩ
M4C MC3301P
UK TEST? M12
NOT POPULATED
MR41 1MΩ
MR2 7.6kΩ
MR3 1kΩ
MC15 472K
VR1 R_P_US
MC1 10µF
VCC_AMP
MC38 104Z
MC4 220µF
M1 SB3731
FEEDBACK BYPASS GND-I GND-O MUTE BS-I IN N.C VCC BS-M OUT-M OUT-I
MC11 104K
MC12 100µF
SPEAKER+
SPEAKER-
MC14 104K
MC13 154J
MR4 10Ω
MR5 10Ω
MR1 100kΩ
MC2 10µF
MC10 100µF
MC3 220µF
AMUTE
+12V
MR13 1.5kΩ
7.7V
MC83 104K
MC25 47µF
MR12 2.7kΩ
MR48 1MΩ
CTLREF
MR49 560kΩ
PTREBLE
M5C MC3301P
1MΩ MR52
472K MC74
PBALD
MR43 560kΩ
CTLREF
MR42 1MΩ
M5B MC3301P
MR47 1MΩ
MC71 472K
MC72 472K
MR46 1MΩ
M5A MC3301P
CTLREF
MR44 1MΩ
PVOL
MR45 560kΩ
472K MC78
1MΩ MR53
M5D MC3301P
1MΩ MR50
CTLREF
MR51 560kΩ
PBASS

For the PSG
Each analog control voltage goes
from 0.55V to 3.7V at the opamp output
Compared to the reference, that makes
from −7.15V (0V at DAC) to −4V (5V at DAC)

MC30 332K
MC31 473K
MC32 104K
MC33 104K
MC80 104K
MR25 1.5kΩ
MC34 47µF
M5E MC3301P
MC36 10µF
MR23 6.8kΩ
1.54V CTLREF
1kΩ MR22
MR24 1.5kΩ
MR30 10kΩ
MR28 10kΩ
M3
MC27 47µF
IN-2
IN-1
MR19 100kΩ
MR21 100kΩ
TONEB
TONEA
MR18 1kΩ
MC29 104K
MC28 104K
MR20 1kΩ
MR29 75kΩ
MR31 75kΩ
104K MC37
OUT-1
+12V
MR27 1.5kΩ
7.7V
MC82 104K
MC35 47µF
MR26 2.7kΩ
BALD
MR57 620kΩ
CTLREF
MR56 1MΩ
M6A MC3301P
MR58 1MΩ
MC77 472K
M6D MC3301P
MR62 1MΩ
MR63 620kΩ
TREBLE
1MΩ MR65
472K MC76
M6C MC3301P
MR60 1MΩ
MR61 620kΩ
BASS
1MΩ MR64
472K MC75
M6B MC3301P
MR54 1MΩ
VOL
1MΩ MR59
472K MC73
MR55 620kΩ

For the MSM
Each analog control voltage goes
from 0.60V to 3.7V at the opamp output
Compared to the reference, that makes
from −7.1V (0V at DAC) to −4V (5V at DAC)

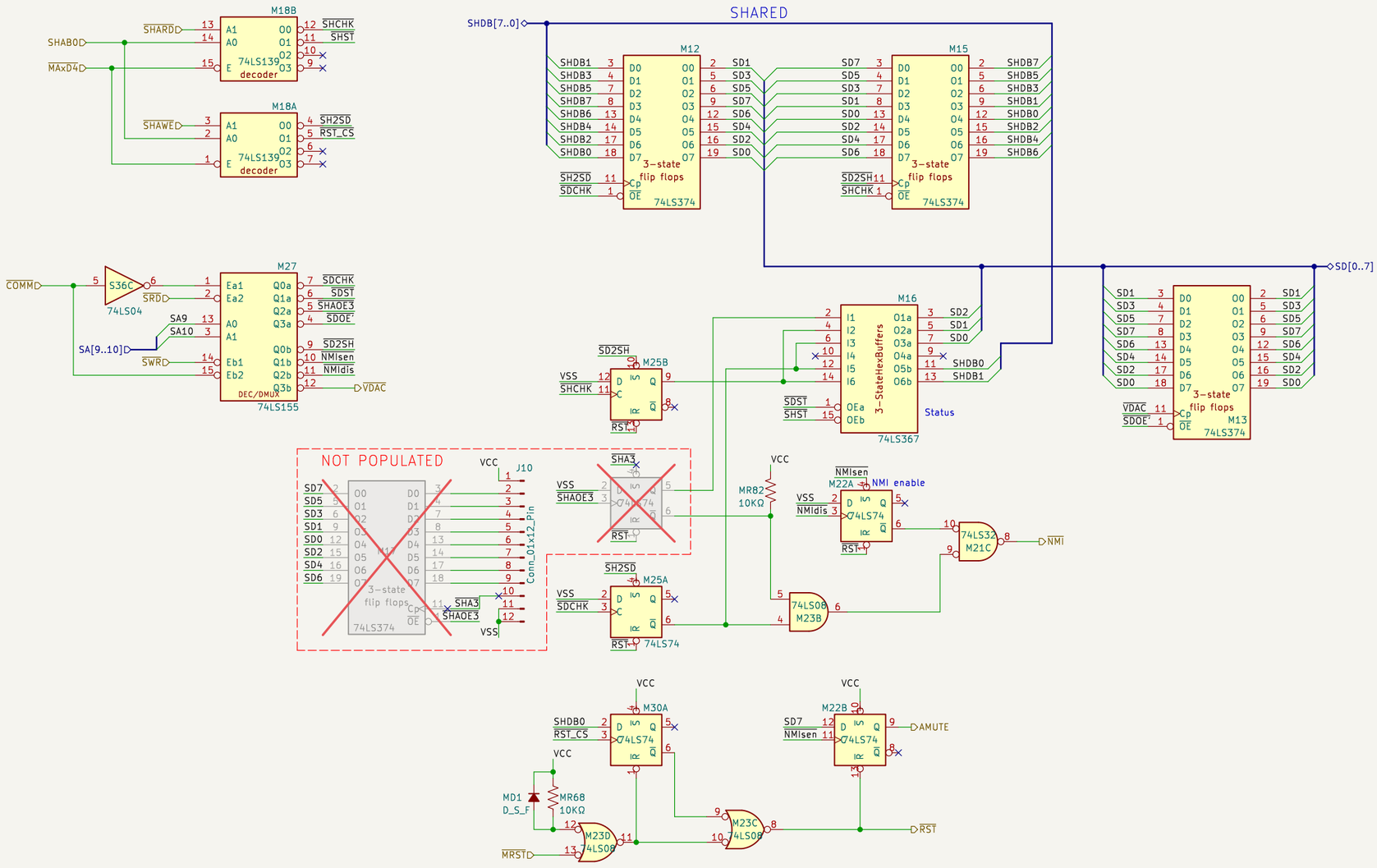
AMUTED
SHAWED
JP2 SJ
PIN 27 SOCKET
NOT USED - B49
SC1
MRST MAxD4 SHARD
SHDB[0..7]
SHDB7 SHDB6 SHDB5 SHDB4 SHDB3 SHDB2 SHDB1 SHDB0
SHAB[0..1]
SHAB1 SHAB0
AVSS
VCC
PWR_FLAG
NO MOUNTED
Conn_01x10_Pin
MN1
L38 L_Small
C6 104Z
MC7 220µF
C8 104Z
MC9 220µF
L40 SPEAKER+
L39 SPEAKER-
C39 103
C40 103

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Sheet: /Sound/Analog/
File: amplifiers.kicad_sch

Title: Analog Signal Processing

Size: A3	Date: 2024-11-16	Rev:
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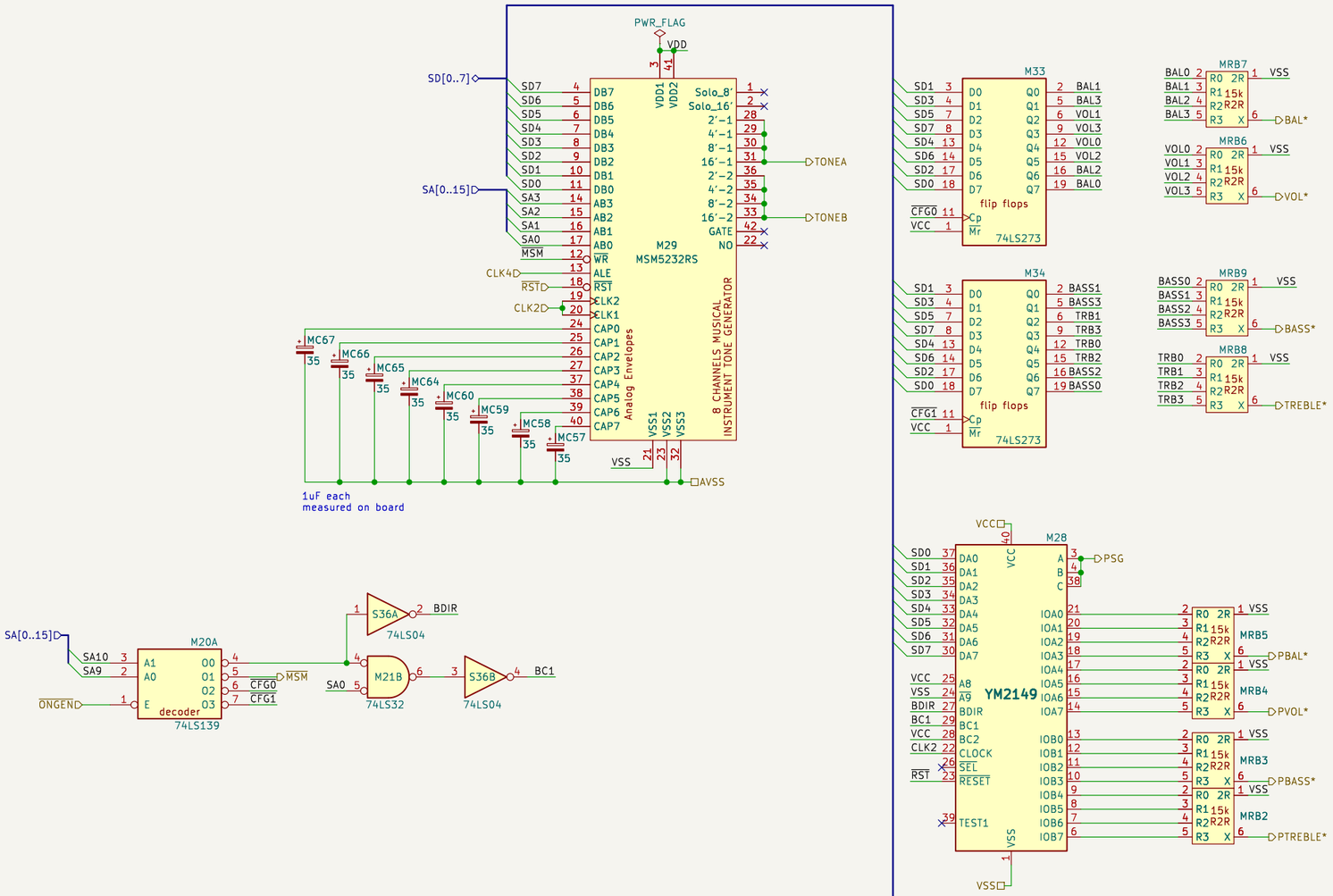


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Sheet: /Sound/Latches/
File: latches.kicad_sch

Title: Communication Latches

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.9		Id: 18/20

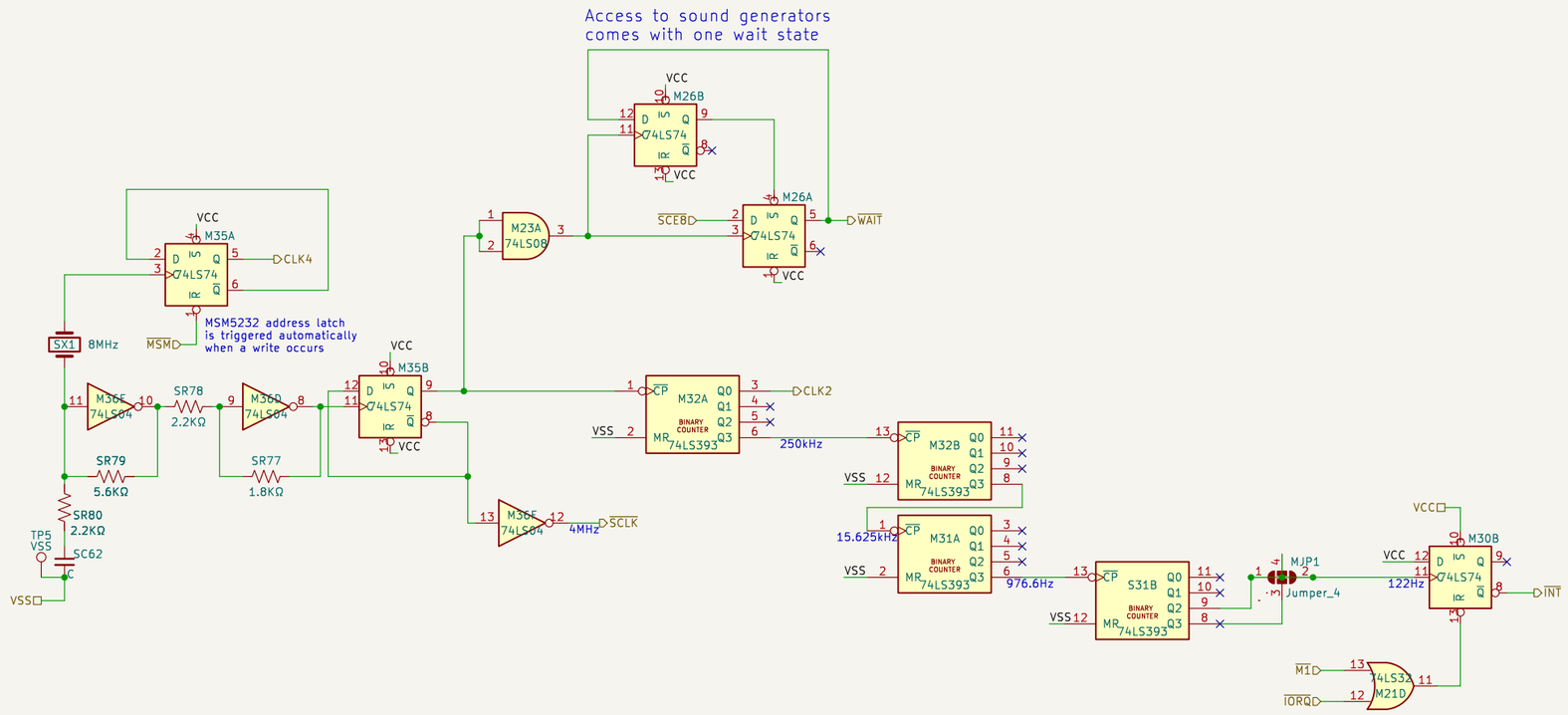


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Sheet: /Sound/Sound Generators/
File: ongen.kicad_sch

Title: Sound Generators

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.9		Id: 19/20



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Sheet: /Sound/Clocks/
File: snd_clocks.kicad_sch

Title: Sound Clock Generation

Size: A3	Date: 2024-11-16	**Rev:**
KiCad E.D.A. 8.0.9		Id: 20/20